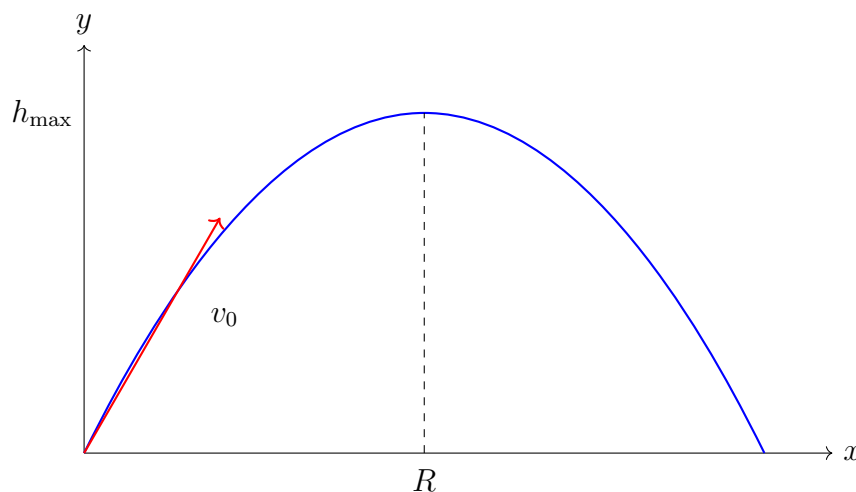


Name: _____ Neptun code: _____

(KTXFI2EBNF) Physics I. Signature Retake Exam Solutions
May 26, 2026

I. Mechanics

1. A projectile is launched at an angle of 60° with an initial velocity of 100 m/s. Calculate:
- (a) The time to reach the peak of the trajectory.
 - (b) The maximum height.
 - (c) The horizontal range (impact distance).
 - (d) The total time of flight.
- ($g = 9.81 \text{ m/s}^2$)



The projectile motion can be separated into horizontal and vertical components. The horizontal velocity remains constant during the motion, while the vertical velocity changes because of gravity.

The horizontal component of the initial velocity is

$$v_{0x} = v_0 \cos 60^\circ = 100 \cdot 0.5 = 50 \text{ m/s}.$$

The vertical component of the initial velocity is

$$v_{0y} = v_0 \sin 60^\circ = 100 \cdot 0.866 = 86.6 \text{ m/s}.$$

(a) Time to reach the highest point

At the highest point of the trajectory the vertical velocity becomes zero. Therefore,

$$0 = v_{0y} - gt_{\text{peak}}.$$

Rearranging the equation gives

$$t_{\text{peak}} = \frac{v_{0y}}{g}.$$

Substituting the numerical values,

$$t_{\text{peak}} = \frac{86.6 \text{ m/s}}{9.81 \text{ m/s}^2} = 8.83 \text{ s}.$$

Therefore,

$$\boxed{t_{\text{peak}} = 8.83 \text{ s}}.$$

(b) Maximum height

The maximum height can be calculated from the kinematic relation

$$h_{\text{max}} = \frac{v_{0y}^2}{2g}.$$

Substituting the values,

$$h_{\text{max}} = \frac{(86.6 \text{ m/s})^2}{2 \cdot 9.81 \text{ m/s}^2} = 382 \text{ m}.$$

Thus,

$$\boxed{h_{\text{max}} = 382 \text{ m}}.$$

(c) Horizontal range

The total flight time is twice the time required to reach the peak:

$$t_{\text{total}} = 2t_{\text{peak}} = 2 \cdot 8.83 \text{ s} = 17.66 \text{ s}.$$

The horizontal range is therefore

$$R = v_{0x} t_{\text{total}}.$$

Substituting the numerical values,

$$R = 50 \text{ m/s} \cdot 17.66 \text{ s} = 883 \text{ m}.$$

Hence,

$$\boxed{R = 883 \text{ m}}.$$

(d) Total time of flight

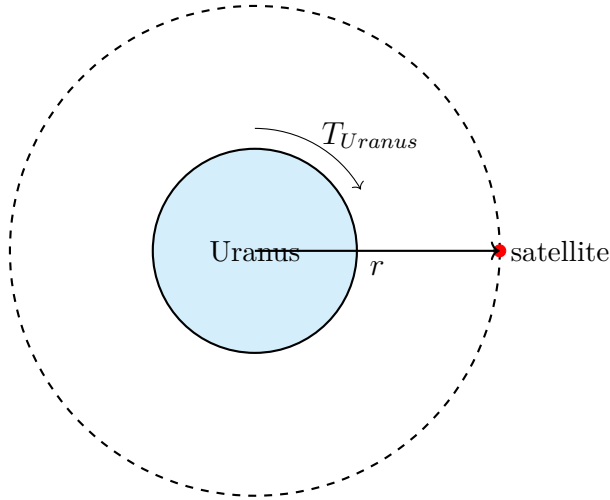
As calculated previously, the total flight time is

$$t_{\text{total}} = 17.66 \text{ s.}$$

Therefore,

$$\boxed{t_{\text{total}} = 17.66 \text{ s.}}$$

2. At what distance from the Uranus does an geosynchronous satellite orbit in the plane of the Equator if it remains constantly above the same point on the Uranus's surface? ($M_{\text{Uranus}} = 8.68 \cdot 10^{25} \text{ kg}$, $T_{\text{Uranus}} = -0,72 \text{ day}$, $R_{\text{Uranus}} = 25300 \text{ km}$, $G = 6.6743 \cdot 10^{-11} \text{ Nm}^2/\text{kg}^2$.)



A geosynchronous satellite must have the same angular velocity as the rotation of Uranus. The negative sign of the given period indicates the direction of rotation, but the distance of the orbit depends on the magnitude of the period.

$$T = 0.72 \text{ day} = 0.72 \cdot 24 \cdot 3600 \text{ s} = 62208 \text{ s.}$$

For a circular orbit, the gravitational force provides the centripetal force:

$$\frac{GMm}{r^2} = m \frac{4\pi^2 r}{T^2}.$$

The satellite mass cancels, and the orbital radius measured from the center of Uranus is

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}.$$

Substituting the data,

$$r = \sqrt[3]{\frac{6.6743 \cdot 10^{-11} \text{ Nm}^2/\text{kg}^2 \cdot 8.68 \cdot 10^{25} \text{ kg} \cdot (62208 \text{ s})^2}{4\pi^2}}.$$

This gives

$$r = 8.28 \cdot 10^7 \text{ m.}$$

This is the distance from the center of Uranus. The distance from the surface is obtained by subtracting the radius of Uranus.

$$R_{Uranus} = 25300 \text{ km} = 2.53 \cdot 10^7 \text{ m.}$$

$$h = r - R_{Uranus} = 8.28 \cdot 10^7 \text{ m} - 2.53 \cdot 10^7 \text{ m} = 5.75 \cdot 10^7 \text{ m.}$$

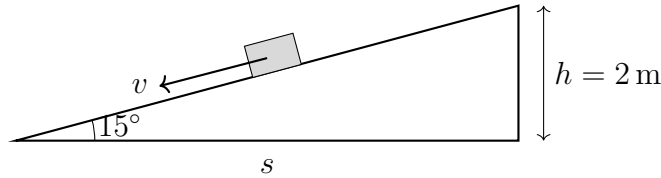
Therefore the satellite orbits at the altitude

$$h = 5.75 \cdot 10^7 \text{ m} = 57500 \text{ km}.$$

The corresponding orbital radius from the center of Uranus is

$$r = 8.28 \cdot 10^7 \text{ m} = 82800 \text{ km}.$$

3. A 5 kg object slides down from the top of a 2-meter-high inclined plane that makes an angle of 15° with the horizontal. What is the velocity of the object when it reaches the bottom of the incline, starting from rest at the top, if:
- (a) there is no friction between the object and the incline?
 - (b) the coefficient of friction is 0.2?
- ($g = 9.81 \text{ m/s}^2$)



First, consider the case without friction. Since the object starts from rest and there is no energy loss, the initial gravitational potential energy is completely converted into kinetic energy at the bottom.

$$mgh = \frac{1}{2}mv^2.$$

The mass cancels from the equation, so the result does not depend on the mass in the frictionless case.

$$gh = \frac{1}{2}v^2.$$

Thus,

$$v = \sqrt{2gh}.$$

Substituting the values,

$$v = \sqrt{2 \cdot 9.81 \text{ m/s}^2 \cdot 2 \text{ m}} = 6.26 \text{ m/s}.$$

Therefore, without friction,

$$\boxed{v = 6.26 \text{ m/s}}.$$

Now consider the case with friction. The length of the inclined plane is needed because the work of friction depends on the distance travelled along the incline.

$$s = \frac{h}{\sin 15^\circ} = \frac{2 \text{ m}}{\sin 15^\circ} = 7.73 \text{ m}.$$

The normal force is

$$N = mg \cos 15^\circ.$$

The friction force is

$$F_f = \mu N = \mu mg \cos 15^\circ.$$

The work done by friction is negative, because friction removes mechanical energy from the system.

$$W_f = -F_f s = -\mu mg \cos 15^\circ s.$$

The energy equation is therefore

$$mgh - \mu mg \cos 15^\circ s = \frac{1}{2}mv^2.$$

Again the mass cancels:

$$gh - \mu g \cos 15^\circ s = \frac{1}{2}v^2.$$

Solving for the velocity gives

$$v = \sqrt{2gh - 2\mu g \cos 15^\circ s}.$$

Substituting the numerical values,

$$v = \sqrt{2 \cdot 9.81 \text{ m/s}^2 \cdot 2 \text{ m} - 2 \cdot 0.2 \cdot 9.81 \text{ m/s}^2 \cdot \cos 15^\circ \cdot 7.73 \text{ m}}.$$

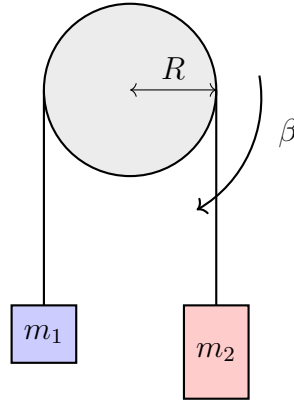
This gives

$$v = 3.15 \text{ m/s}.$$

Therefore, with friction,

$$v = 3.15 \text{ m/s}.$$

4. A string is wound around a solid cylinder with radius $R = 0.6 \text{ m}$ and mass $M = 50 \text{ kg}$. Two objects with a combined mass of 25 kg are hanging from the ends of the string. Upon release, the cylinder accelerates with an angular acceleration of $\beta = 5 \text{ rad/s}^2$. The string does not slip. ($g = 9.81 \text{ m/s}^2$)
- Find the individual masses (m_1, m_2) of the two objects.
 - Calculate the total kinetic energy of the system at $t = 5 \text{ s}$.



Since the string does not slip, the linear acceleration of the hanging masses is related to the angular acceleration of the cylinder by

$$a = R\beta.$$

Substituting the data,

$$a = 0.6 \text{ m} \cdot 5 \text{ rad/s}^2 = 3.0 \text{ m/s}^2.$$

Let m_2 be the heavier mass, so that m_2 moves downward and m_1 moves upward. The equations of motion for the two hanging masses are

$$m_2g - T_2 = m_2a.$$

$$T_1 - m_1g = m_1a.$$

The cylinder is a solid cylinder, so its moment of inertia is

$$I = \frac{1}{2}MR^2.$$

The net torque acting on the cylinder is caused by the two string tensions:

$$(T_2 - T_1)R = I\beta.$$

Substituting $I = \frac{1}{2}MR^2$ and $a = R\beta$, we get

$$T_2 - T_1 = \frac{1}{2}Ma.$$

From the two translational equations,

$$T_2 = m_2g - m_2a = m_2(g - a).$$

$$T_1 = m_1g + m_1a = m_1(g + a).$$

Therefore,

$$m_2(g - a) - m_1(g + a) = \frac{1}{2}Ma.$$

We also know that

$$m_1 + m_2 = 25 \text{ kg}.$$

Substituting the numerical values into the torque equation,

$$m_2(9.81 - 3.00) - m_1(9.81 + 3.00) = \frac{1}{2} \cdot 50 \cdot 3.00.$$

Thus,

$$6.81m_2 - 12.81m_1 = 75.0.$$

Using $m_2 = 25 - m_1$,

$$6.81(25 - m_1) - 12.81m_1 = 75.0.$$

Solving,

$$170.25 - 19.62m_1 = 75.0.$$

$$m_1 = 4.86 \text{ kg}.$$

Then

$$m_2 = 25.0 \text{ kg} - 4.86 \text{ kg} = 20.14 \text{ kg}.$$

Thus the individual masses are

$$\boxed{m_1 = 4.86 \text{ kg}}, \quad \boxed{m_2 = 20.14 \text{ kg}}.$$

Now calculate the kinetic energy at $t = 5 \text{ s}$. Since the system starts from rest and the acceleration is constant,

$$v = at = 3.0 \text{ m/s}^2 \cdot 5 \text{ s} = 15.0 \text{ m/s}.$$

The angular velocity of the cylinder is

$$\omega = \beta t = 5 \text{ rad/s}^2 \cdot 5 \text{ s} = 25 \text{ rad/s}.$$

The translational kinetic energy of the two masses is

$$K_{\text{trans}} = \frac{1}{2}(m_1 + m_2)v^2.$$

Substituting,

$$K_{\text{trans}} = \frac{1}{2} \cdot 25 \text{ kg} \cdot (15.0 \text{ m/s})^2 = 2812.5 \text{ J}.$$

The rotational kinetic energy of the cylinder is

$$K_{\text{rot}} = \frac{1}{2}I\omega^2.$$

The moment of inertia is

$$I = \frac{1}{2}MR^2 = \frac{1}{2} \cdot 50 \text{ kg} \cdot (0.6 \text{ m})^2 = 9.0 \text{ kg m}^2.$$

Therefore,

$$K_{\text{rot}} = \frac{1}{2} \cdot 9.0 \text{ kg m}^2 \cdot (25 \text{ rad/s})^2 = 2812.5 \text{ J}.$$

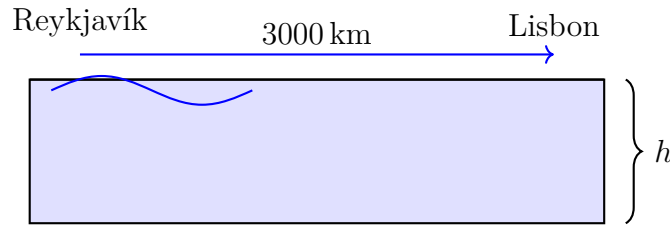
The total kinetic energy is

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = 2812.5 \text{ J} + 2812.5 \text{ J} = 5625 \text{ J}.$$

Thus,

$$\boxed{K_{\text{total}} = 5.63 \cdot 10^3 \text{ J}}.$$

5. Estimate the average depth of the Atlantic Ocean if a tsunami takes four hours to travel from Reykjavík to Lisbon. (Note: The propagation speed of shallow-water waves is $v = \sqrt{gh}$, where $g = 9.81 \text{ m/s}^2$. The distance is 3000 km.)



The tsunami is treated as a shallow-water wave. The propagation speed of such a wave is given by

$$v = \sqrt{gh}.$$

The travelled distance between Reykjavík and Lisbon is

$$s = 3000 \text{ km} = 3.00 \cdot 10^6 \text{ m}.$$

The travel time is now

$$t = 4 \text{ h} = 4 \cdot 3600 \text{ s} = 14400 \text{ s}.$$

The average propagation speed of the tsunami is therefore

$$v = \frac{s}{t}.$$

Substituting the numerical values,

$$v = \frac{3.00 \cdot 10^6 \text{ m}}{14400 \text{ s}} = 208.3 \text{ m/s}.$$

For shallow-water waves,

$$v = \sqrt{gh}.$$

Squaring both sides gives

$$v^2 = gh.$$

Thus the average ocean depth is

$$h = \frac{v^2}{g}.$$

Substituting the values,

$$h = \frac{(208.3 \text{ m/s})^2}{9.81 \text{ m/s}^2} = 4424 \text{ m}.$$

Therefore the estimated average depth of the Atlantic Ocean is

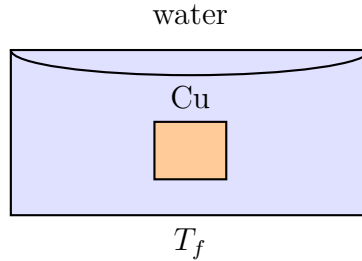
$$h \approx 4.42 \cdot 10^3 \text{ m}.$$

So the estimated average depth is approximately

$$h \approx 4.4 \text{ km}.$$

II. Thermodynamics

6. We place a copper piece with a mass of 500 gramms and a temperature of 40°C into 1250 gramms of water at 70°C . What will be the common temperature? The specific heat capacity of copper is $3.85 \cdot 10^2 \frac{\text{J}}{\text{kg K}}$ and the specific heat capacity of water is $4.19 \cdot 10^3 \frac{\text{J}}{\text{kg K}}$.



The water is initially hotter than the copper block, therefore thermal energy flows from the water to the copper until thermal equilibrium is reached. We assume that the system is thermally insulated, so no heat is exchanged with the environment.

The heat lost by the water is equal to the heat gained by the copper:

$$Q_{\text{water}} = Q_{\text{copper}}.$$

Using the calorimetric equation,

$$m_w c_w (T_w - T_f) = m_{Cu} c_{Cu} (T_f - T_{Cu}).$$

First we convert all quantities into SI units.

The mass of the copper block is

$$m_{Cu} = 500 \text{ g} = 0.500 \text{ kg}.$$

The mass of the water is

$$m_w = 1250 \text{ g} = 1.250 \text{ kg}.$$

The specific heat capacities are already given in SI units:

$$c_{Cu} = 3.85 \cdot 10^2 \text{ J/(kg K)}.$$

$$c_w = 4.19 \cdot 10^3 \text{ J/(kg K)}.$$

Substituting all numerical values into the energy balance equation,

$$1.250 \text{ kg} \cdot 4.19 \cdot 10^3 \text{ J/(kg K)} \cdot (70 - T_f) = 0.500 \text{ kg} \cdot 3.85 \cdot 10^2 \text{ J/(kg K)} \cdot (T_f - 40).$$

Expanding the equation,

$$5237.5(70 - T_f) = 192.5(T_f - 40).$$

After rearranging,

$$366625 - 5237.5T_f = 192.5T_f - 7700.$$

$$374325 = 5430T_f.$$

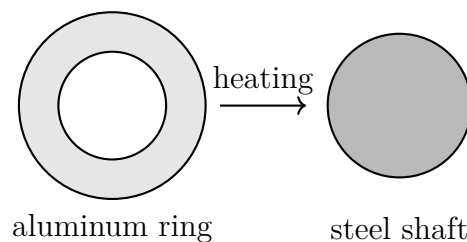
Thus,

$$T_f = 68.94^\circ\text{C}.$$

Therefore the equilibrium temperature of the system is

$$\boxed{T_f = 68.9^\circ\text{C}}.$$

7. At a temperature of 50°C , a steel shaft with a diameter of 11.21 cm needs to be fitted with an aluminum ring with an inner diameter of 11.18 cm at the same temperature. The linear thermal expansion coefficient of steel is $12 \cdot 10^{-6} \frac{1}{\text{K}}$, and the linear thermal expansion coefficient of aluminum is $28.7 \cdot 10^{-6} \frac{1}{\text{K}}$. What temperature should the ring be heated to?



The aluminum ring must expand sufficiently so that its inner diameter becomes equal to the diameter of the steel shaft.

The initial diameter of the aluminum ring is

$$d_0 = 11.18 \text{ cm} = 0.1118 \text{ m}.$$

The required final diameter is

$$d = 11.21 \text{ cm} = 0.1121 \text{ m}.$$

The linear thermal expansion formula is

$$d = d_0(1 + \alpha\Delta T).$$

The expansion coefficient of aluminum is

$$\alpha = 28.7 \cdot 10^{-6} \text{ K}^{-1}.$$

The increase in diameter is

$$\Delta d = d - d_0 = 0.1121 \text{ m} - 0.1118 \text{ m} = 3.0 \cdot 10^{-4} \text{ m}.$$

Using the linear expansion equation,

$$\Delta d = d_0\alpha\Delta T.$$

Therefore,

$$\Delta T = \frac{\Delta d}{d_0\alpha}.$$

Substituting the numerical values,

$$\Delta T = \frac{3.0 \cdot 10^{-4} \text{ m}}{0.1118 \text{ m} \cdot 28.7 \cdot 10^{-6} \text{ K}^{-1}} = 93.5 \text{ K}.$$

The initial temperature of the ring is

$$T_0 = 50^\circ\text{C}.$$

Thus the required final temperature is

$$T = T_0 + \Delta T = 50^\circ\text{C} + 93.5^\circ\text{C} = 143.5^\circ\text{C}.$$

Therefore the aluminum ring should be heated to approximately

$$\boxed{T = 144^\circ\text{C}}.$$

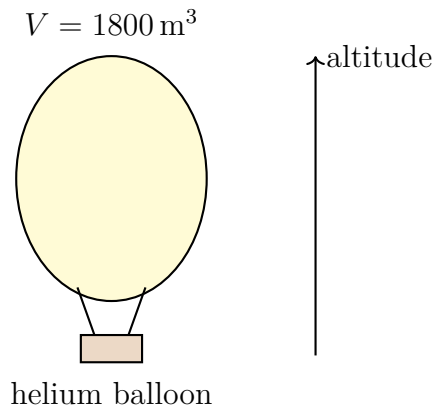
8. The atmospheric balloon floating at an altitude of several kilometers has a volume of 1800 m^3 . The temperature and pressure of the helium inside it are the same as the surrounding -40°C and $5 \cdot 10^3\text{ Pa}$, respectively.

(a) How many moles of helium are in the balloon?

(b) What is the mass of the helium gas? ($M = 4\frac{\text{g}}{\text{mol}}$)

(c) What was the volume of the balloon at the time of launch on Earth, if the gas inside the balloon was characterized by 0°C and 10^5 Pa pressure?

$$(R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}})$$



The helium gas inside the balloon can be treated as an ideal gas. Therefore we use the ideal gas equation

$$pV = nRT.$$

The pressure is

$$p = 5.0 \cdot 10^3\text{ Pa}.$$

The volume is

$$V = 1800\text{ m}^3.$$

The temperature must be converted into kelvin:

$$T = -40^\circ\text{C} + 273.15 = 233.15\text{ K}.$$

(a) Number of moles

Rearranging the ideal gas equation,

$$n = \frac{pV}{RT}.$$

Substituting the numerical values,

$$n = \frac{5.0 \cdot 10^3 \text{ Pa} \cdot 1800 \text{ m}^3}{8.31 \text{ J}/(\text{mol K}) \cdot 273.15 \text{ K}}.$$

This gives

$$n = 4.65 \cdot 10^3 \text{ mol}.$$

Therefore the amount of helium inside the balloon is

$$\boxed{n = 4.65 \cdot 10^3 \text{ mol}}.$$

(b) Mass of helium

The molar mass of helium is

$$M = 4 \text{ g/mol} = 4.0 \cdot 10^{-3} \text{ kg/mol}.$$

The mass of the helium is

$$m = nM.$$

Substituting the numerical values,

$$m = 4.65 \cdot 10^3 \text{ mol} \cdot 4.0 \cdot 10^{-3} \text{ kg/mol} = 18.6 \text{ kg}.$$

Thus,

$$\boxed{m = 18.6 \text{ kg}}.$$

(c) Volume at launch

At launch conditions,

$$T_0 = 0^\circ\text{C} + 273.15 = 273.15 \text{ K}.$$

$$p_0 = 1.0 \cdot 10^5 \text{ Pa}.$$

The amount of helium remains constant, therefore

$$V_0 = \frac{nRT_0}{p_0}.$$

Substituting the values,

$$V_0 = \frac{4.65 \cdot 10^3 \text{ mol} \cdot 8.31 \text{ J}/(\text{mol K}) \cdot 273.15 \text{ K}}{1.0 \cdot 10^5 \text{ Pa}}.$$

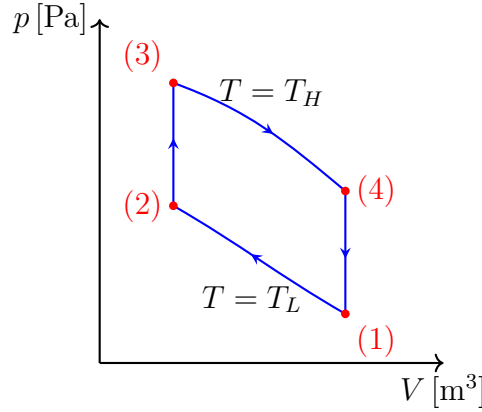
This gives

$$V_0 = 105 \text{ m}^3.$$

Therefore the initial volume of the balloon at launch was

$$\boxed{V_0 = 105 \text{ m}^3}.$$

9. The ideal cycle of an externally heated heat engine, shown in the diagram, differs from the Otto engine cycle in that it contains isothermal processes instead of adiabatic ones.



What is the efficiency of the cycle if $p_3 = 2p_2$, the compression ratio is 8, and the cycle is performed with a two-atomic (diatomic) gas? What is the efficiency of the Carnot cycle taking place between the same temperature limits? ($R = 8.31 \text{ J}/(\text{mol K})$)

The thermodynamic cycle consists of two isothermal processes and two isochoric processes.

The processes are:

- $1 \rightarrow 2$: isothermal compression,
- $2 \rightarrow 3$: isochoric heating,
- $3 \rightarrow 4$: isothermal expansion,
- $4 \rightarrow 1$: isochoric cooling.

Because the gas is ideal, the temperature is proportional to the product pV :

$$T \propto pV.$$

During the isochoric process $2 \rightarrow 3$, the volume remains constant. Therefore,

$$\frac{T_3}{T_2} = \frac{p_3}{p_2}.$$

The problem states that

$$p_3 = 2p_2.$$

Thus,

$$\frac{T_3}{T_2} = 2.$$

Therefore the higher temperature is twice the lower temperature:

$$T_H = T_3 = T_4 = 2T_2 = 2T_L.$$

The compression ratio is defined as

$$r = \frac{V_1}{V_2} = 8.$$

For a diatomic ideal gas, the molar heat capacity at constant volume is

$$C_V = \frac{5}{2}R.$$

Net work of the cycle

During an isothermal process, the work done by the gas is

$$W = nRT \ln \frac{V_f}{V_i}.$$

For the isothermal compression process $1 \rightarrow 2$,

$$W_{12} = nRT_L \ln \frac{V_2}{V_1} = -nRT_L \ln r.$$

For the isothermal expansion process $3 \rightarrow 4$,

$$W_{34} = nRT_H \ln \frac{V_4}{V_3} = nRT_H \ln r.$$

The isochoric processes do not perform mechanical work because the volume does not change.

Therefore the total work of the cycle is

$$W = W_{12} + W_{34} = nR(T_H - T_L) \ln r.$$

Since

$$T_H = 2T_L,$$

we obtain

$$W = nRT_L \ln r.$$

Heat absorbed during the cycle

Heat enters the system during the isochoric heating process and during the high-temperature isothermal expansion.

For the isochoric heating process,

$$Q_{23} = nC_V(T_H - T_L).$$

Using $T_H = 2T_L$,

$$Q_{23} = nC_V T_L.$$

For the isothermal expansion process,

$$Q_{34} = nRT_H \ln r.$$

Thus the total absorbed heat is

$$Q_{\text{in}} = nC_V T_L + nRT_H \ln r.$$

Again using $T_H = 2T_L$,

$$Q_{\text{in}} = nC_V T_L + 2nRT_L \ln r.$$

Efficiency of the engine

The thermal efficiency is defined as

$$\eta = \frac{W}{Q_{\text{in}}}.$$

Substituting the expressions derived above,

$$\eta = \frac{nRT_L \ln r}{nC_V T_L + 2nRT_L \ln r}.$$

After cancellation of the common factor nT_L ,

$$\eta = \frac{R \ln r}{C_V + 2R \ln r}.$$

Using

$$C_V = \frac{5}{2}R,$$

the efficiency becomes

$$\eta = \frac{\ln r}{\frac{5}{2} + 2 \ln r}.$$

Since

$$r = 8,$$

we have

$$\ln 8 = 2.079.$$

Therefore,

$$\eta = \frac{2.079}{2.5 + 2 \cdot 2.079} = 0.312.$$

Thus the efficiency of this heat engine cycle is

$$\boxed{\eta = 0.312 = 31.2\%}.$$

Carnot efficiency

The Carnot efficiency between the same temperature limits is

$$\eta_C = 1 - \frac{T_L}{T_H}.$$

Since

$$T_H = 2T_L,$$

we obtain

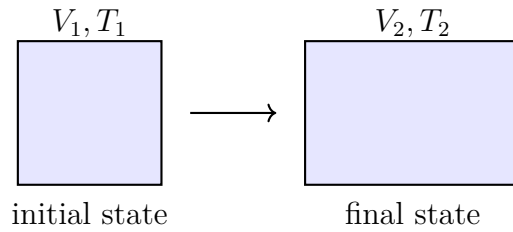
$$\eta_C = 1 - \frac{1}{2} = 0.5.$$

Therefore the efficiency of the corresponding Carnot cycle is

$$\boxed{\eta_C = 0.500 = 50.0\%}.$$

The Carnot cycle is more efficient because all heat transfer occurs reversibly at constant temperature, while in the present cycle a part of the heat transfer takes place during isochoric heating, which is thermodynamically less efficient.

10. What is the entropy change if the volume of $m = 2$ g of hydrogen changes from $V_1 = 30$ l to $V_2 = 120$ l, and its temperature changes from $t_1 = 80^\circ\text{C}$ to $t_2 = 400^\circ\text{C}$? The molar mass is $M = 2$ g/mol, $R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}}$.



The entropy change of an ideal gas between two equilibrium states can be calculated from

$$\Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}.$$

Hydrogen is a diatomic gas under normal conditions, therefore

$$C_V = \frac{5}{2}R.$$

The amount of substance is

$$n = \frac{m}{M}.$$

The given mass is

$$m = 2 \text{ g} = 2.0 \cdot 10^{-3} \text{ kg}.$$

The molar mass is

$$M = 2 \text{ g/mol} = 2.0 \cdot 10^{-3} \text{ kg/mol}.$$

Thus,

$$n = \frac{2.0 \cdot 10^{-3} \text{ kg}}{2.0 \cdot 10^{-3} \text{ kg/mol}} = 1.0 \text{ mol}.$$

The temperatures in kelvin are

$$T_1 = 80^\circ\text{C} + 273.15 = 353.15 \text{ K}.$$

$$T_2 = 400^\circ\text{C} + 273.15 = 673.15 \text{ K}.$$

The initial and final volumes converted into SI units are

$$V_1 = 301 = 3.0 \cdot 10^{-2} \text{ m}^3.$$

$$V_2 = 1201 = 1.20 \cdot 10^{-1} \text{ m}^3.$$

The volume ratio is therefore

$$\frac{V_2}{V_1} = 4.$$

Substituting all quantities into the entropy equation,

$$\Delta S = 1.0 \text{ mol} \cdot \frac{5}{2} \cdot 8.31 \text{ J}/(\text{mol K}) \cdot \ln \frac{673.15 \text{ K}}{353.15 \text{ K}} + 1.0 \text{ mol} \cdot 8.31 \text{ J}/(\text{mol K}) \cdot \ln 4.$$

The temperature contribution is

$$\Delta S_T = 13.4 \text{ J/K}.$$

The volume contribution is

$$\Delta S_V = 11.5 \text{ J/K}.$$

Therefore the total entropy change is

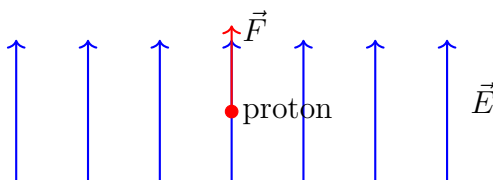
$$\Delta S = 13.4 \text{ J/K} + 11.5 \text{ J/K} = 24.9 \text{ J/K}.$$

Thus the entropy increase of the hydrogen gas is

$$\boxed{\Delta S = 24.9 \text{ J/K}}.$$

III. Electrostatics, Waves, and Optics

11. Determine the acceleration of a proton $m = 1.67 \cdot 10^{-27} \text{ kg}$ in an electric field of strength $E = 300 \text{ N/C}$. How many times greater is this acceleration than the acceleration due to gravity? ($e = 1.6 \cdot 10^{-19} \text{ C}$, $g = 9.81 \text{ m/s}^2$)



A proton placed in an electric field experiences an electric force. The magnitude of the electric force acting on a charged particle is

$$F = qE.$$

For a proton, the electric charge is

$$q = e = 1.6 \cdot 10^{-19} \text{ C}.$$

The electric field strength is

$$E = 300 \text{ N/C}.$$

Therefore the electric force acting on the proton is

$$F = (1.6 \cdot 10^{-19} \text{ C})(300 \text{ N/C}) = 4.8 \cdot 10^{-17} \text{ N}.$$

According to Newton's second law,

$$F = ma.$$

Thus the acceleration is

$$a = \frac{F}{m}.$$

The proton mass is

$$m = 1.67 \cdot 10^{-27} \text{ kg}.$$

Substituting the numerical values,

$$a = \frac{4.8 \cdot 10^{-17} \text{ N}}{1.67 \cdot 10^{-27} \text{ kg}} = 2.87 \cdot 10^{10} \text{ m/s}^2.$$

Therefore the acceleration of the proton is

$$\boxed{a = 2.87 \cdot 10^{10} \text{ m/s}^2}.$$

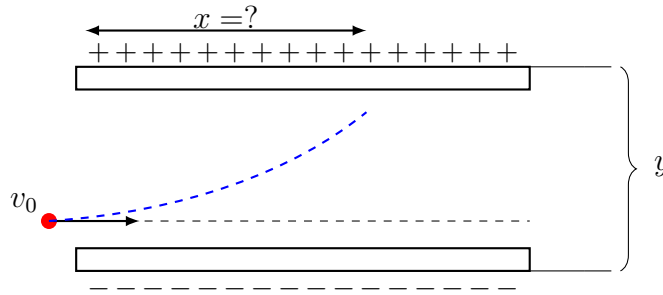
Now compare this acceleration to the acceleration due to gravity.

The ratio is

$$\frac{a}{g} = \frac{2.87 \cdot 10^{10} \text{ m/s}^2}{9.81 \text{ m/s}^2} = 2.93 \cdot 10^9.$$

Thus the electric acceleration is approximately $\boxed{2.93 \cdot 10^9}$ times greater than the gravitational acceleration.

12. An electron is launched with an initial velocity of $v_0 = 3 \cdot 10^6$ m/s between two parallel charged plates, as shown in the figure. If the electric field strength between the plates is $E = 3$ kN/C, where does the electron hit the upper plate? Assume that the arrangement is in vacuum.



$$(m_e = 9.1 \cdot 10^{-31} \text{ kg}, e = 1.6 \cdot 10^{-19} \text{ C})$$

The electron moves horizontally with constant velocity while simultaneously accelerating vertically because of the electric field. The motion is therefore analogous to projectile motion.

The electric field strength is

$$E = 3 \text{ kN/C} = 3.0 \cdot 10^3 \text{ N/C}.$$

The electric force acting on the electron is

$$F = eE.$$

Substituting the values,

$$F = (1.6 \cdot 10^{-19} \text{ C})(3.0 \cdot 10^3 \text{ N/C}) = 4.8 \cdot 10^{-16} \text{ N}.$$

The acceleration of the electron is

$$a = \frac{F}{m_e}.$$

Using the electron mass,

$$m_e = 9.1 \cdot 10^{-31} \text{ kg},$$

we obtain

$$a = \frac{4.8 \cdot 10^{-16} \text{ N}}{9.1 \cdot 10^{-31} \text{ kg}} = 5.27 \cdot 10^{14} \text{ m/s}^2.$$

The electron starts halfway between the plates. The plate separation from the figure is

$$y = 2.0 \text{ cm} = 2.0 \cdot 10^{-2} \text{ m}.$$

Therefore the vertical distance travelled before hitting the upper plate is

$$s_y = \frac{y}{2} = 1.0 \cdot 10^{-2} \text{ m.}$$

The initial vertical velocity is zero, so the vertical motion equation is

$$s_y = \frac{1}{2}at^2.$$

Solving for the time,

$$t = \sqrt{\frac{2s_y}{a}}.$$

Substituting the numerical values,

$$t = \sqrt{\frac{2 \cdot 1.0 \cdot 10^{-2} \text{ m}}{5.27 \cdot 10^{14} \text{ m/s}^2}} = 6.16 \cdot 10^{-9} \text{ s.}$$

The horizontal velocity remains constant:

$$v_0 = 3.0 \cdot 10^6 \text{ m/s.}$$

Thus the horizontal distance travelled before collision is

$$x = v_0 t.$$

Substituting,

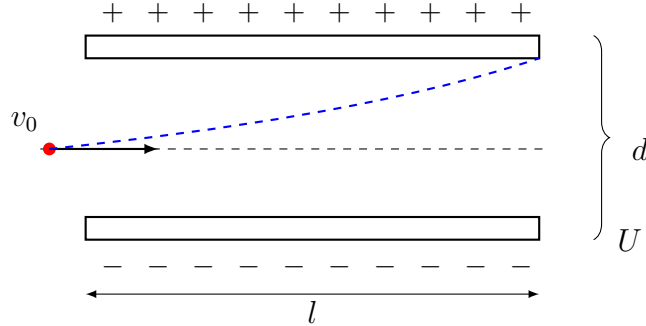
$$x = (3.0 \cdot 10^6 \text{ m/s})(6.16 \cdot 10^{-9} \text{ s}) = 1.85 \cdot 10^{-2} \text{ m.}$$

Therefore the electron hits the upper plate at the distance

$$\boxed{x = 1.85 \cdot 10^{-2} \text{ m} = 1.85 \text{ cm}}$$

from the entrance point.

13. An electron enters the region between the plate pair shown in the figure halfway between the plates. The length of the plates is $l = 20 \text{ cm}$, their separation is $d = 2 \text{ cm}$, and the voltage between them is $U = 500 \text{ V}$. What must the initial velocity of the electron be if we want it to just barely leave the region between the plates? ($m_e = 9.1 \cdot 10^{-31} \text{ kg}$, $e = 1.6 \cdot 10^{-19} \text{ C}$)



The electron enters midway between the plates and must just barely leave the capacitor region without touching the upper plate. Therefore, during the motion through the plates, the vertical displacement of the electron is equal to half of the plate separation. The plate separation is

$$d = 2.0 \text{ cm} = 2.0 \cdot 10^{-2} \text{ m}.$$

Thus the allowed vertical displacement is

$$s_y = \frac{d}{2} = 1.0 \cdot 10^{-2} \text{ m}.$$

The length of the plates is

$$l = 20 \text{ cm} = 2.0 \cdot 10^{-1} \text{ m}.$$

The electric field strength between parallel plates is

$$E = \frac{U}{d}.$$

Substituting the values,

$$E = \frac{500 \text{ V}}{2.0 \cdot 10^{-2} \text{ m}} = 2.5 \cdot 10^4 \text{ N/C}.$$

The electric force acting on the electron is

$$F = eE.$$

Thus,

$$F = (1.6 \cdot 10^{-19} \text{ C})(2.5 \cdot 10^4 \text{ N/C}) = 4.0 \cdot 10^{-15} \text{ N}.$$

The vertical acceleration is

$$a = \frac{F}{m_e}.$$

Substituting the electron mass,

$$a = \frac{4.0 \cdot 10^{-15} \text{ N}}{9.1 \cdot 10^{-31} \text{ kg}} = 4.40 \cdot 10^{15} \text{ m/s}^2.$$

The initial vertical velocity is zero, therefore the vertical displacement is

$$s_y = \frac{1}{2}at^2.$$

Solving for the travelling time,

$$t = \sqrt{\frac{2s_y}{a}}.$$

Substituting the values,

$$t = \sqrt{\frac{2 \cdot 1.0 \cdot 10^{-2} \text{ m}}{4.40 \cdot 10^{15} \text{ m/s}^2}} = 2.13 \cdot 10^{-9} \text{ s}.$$

The horizontal velocity remains constant, therefore

$$v_0 = \frac{l}{t}.$$

Substituting the numerical values,

$$v_0 = \frac{2.0 \cdot 10^{-1} \text{ m}}{2.13 \cdot 10^{-9} \text{ s}} = 9.39 \cdot 10^7 \text{ m/s}.$$

Therefore the required initial velocity is

$$\boxed{v_0 = 9.4 \cdot 10^7 \text{ m/s}}.$$